Interpretable Bankruptcy Prediction for B2B Trade Credit Risk*

YoungBo Go and Hyunchul Ahn†
Graduate School of Business IT, Kookmin University
{gyb357,hcahn}@kookmin.ac.kr

Abstract

This study builds on a benchmark Multi-Head LSTM (MH-LSTM) architecture for corporate bankruptcy prediction and extends it to the context of B2B trade credit screening by providing interpretable evidence from financial time-series data. Each financial variable is assigned an independent LSTM head to learn its unique temporal dynamics, while multi-period inputs spanning 3, 4, and 5 years capture the cumulative progression of financial distress. Using a benchmark dataset of 8,262 U.S. listed companies with 78,682 observations (1999–2018), MH-LSTM achieves stable performance of ROC-AUC 0.886–0.895 and PR-AUC 0.345–0.407, with bankruptcy detection rate improving monotonically as the window extends. Expected Gradients SHAP and Layer Gradient×Activation analyses consistently identify market_value and net_income as the top predictors, and reveal that recent fiscal-year changes contribute more strongly to predictions than historical records. The Spearman rank correlation of ρ ranging from 0.851 (W3) to 0.895 (W5) across window settings between the two XAI approaches confirms high internal consistency of the interpretations.

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† Corresponding author

1 Introduction

The acceleration of digital transformation and the proliferation of e-commerce platforms are fundamentally reshaping Business-to-Business (B2B) transactions. Clark and Lee [1] noted that B2B e-commerce systems deepen interdependencies among independent firms, requiring tight inter-organizational integration and coordination for improved business performance. In particular, trade credit serves as the most important short-term external financing source for companies [2], making early detection of counterparty financial distress a core managerial competency for supply chain participants.

Empirical evidence on the cascading effects of corporate distress underscores the structural severity of this problem. Hertzel et al. [3] documented significant stock price declines for suppliers of bankrupt firms, with shocks propagating broadly along supply chains. Lian [4] confirmed that a major customer's financial distress positively correlates with supplier distress probability, persisting for up to two years after the distress event.

Supply chain finance (SCF) research has proposed combining behavioral transaction data and network information with financial ratios [5], [6], but these approaches are limited to firms with established transaction histories. Publicly disclosed financial statements, by contrast, are a universally accessible standardized information source that can function independently at the new-counterparty screening stage. Since financial distress is a dynamic process that unfolds gradually over multiple years, time-series models processing annual statements as sequential inputs are theoretically better suited for early warning than single-snapshot approaches.

Bankruptcy prediction research has evolved from traditional statistical models [7], [8] through machine learning to deep learning approaches. Despite this progress, most studies rely on single-period financial snapshots, failing to capture the dynamic accumulation of distress over time. LSTM-based models addressed this by processing sequential annual statements, and Pellegrino et al. [9] further proposed a multi-head structure assigning independent LSTM heads to each financial variable. However, quantifying head-level attribution through XAI methods in such architectures remains an open problem. This study addresses this gap by applying Expected Gradients SHAP and Layer Gradient×Activation to provide interpretable variable-level and temporal attribution of predictions.

Building on the benchmark dataset and MH-LSTM architecture of Pellegrino et al. [9], this study makes three incremental but meaningful contributions. First, it provides head-level and variable-level attribution evidence for MH-LSTM bankruptcy prediction using Expected Gradients SHAP and Layer Gradient×Activation. Second, it examines how attribution patterns change across 3-, 4-, and 5-year financial windows. Third, it evaluates the internal consistency of two complementary XAI perspectives by comparing input-space SHAP importance with projection-layer Grad×Act importance.

2 Related Work

2.1 Credit Risk Prediction in B2B Supply Chain Environments

Credit risk prediction in supply chain contexts requires understanding inter-firm dependencies beyond individual financial health. Zhang et al. [5] demonstrated that behavioral transaction data from trading platforms can signal financial distress before it appears in financial statements, achieving significantly higher prediction accuracy than financial ratios alone. Yao et al. [6] addressed class imbalance in supply chain credit prediction by combining differential sampling, SMOTE, and AdaBoost within a decision-tree ensemble framework.

2.2 Traditional Statistical Models

Bankruptcy prediction research originates with Beaver's [7] univariate analysis and Altman's [8] multivariate discriminant Z-Score model. Despite its landmark status, Altman's 50:50 matched-sample design introduces sample selection bias. Ohlson [10] advanced the methodology with logistic regression on an unmatched sample, enabling direct probabilistic interpretation of the output. Zmijewski [11] rigorously demonstrated that oversampling bankrupt firms in training data introduces asymptotic bias into parameter estimates, establishing stratified sampling and imbalance correction as methodological standards.

2.3 Machine Learning and Deep Learning Models

Barboza et al. [12] conducted a systematic comparison of SVM, bagging, boosting, random forests, and traditional statistical models, finding that ensemble methods achieved approximately 10 percentage points higher accuracy than traditional approaches. Dasilas and Rigani [13] confirmed that gradient boosting models (XGBoost, LightGBM) serve as strong baselines across diverse datasets. du Jardin [14] demonstrated that explicitly incorporating period-to-period changes in financial ratios into a dynamic model significantly improves both prediction accuracy and model stability relative to static cross-sectional approaches.

Kim et al. [15] confirmed that LSTM consistently outperforms traditional classifiers on AUC in sequential input settings combining quarterly accounting and daily market data. However, the standard single-head LSTM conflates heterogeneous variable dynamics into a single hidden state. Pellegrino et al. [9] proposed a multi-head architecture assigning independent LSTM heads per variable and publicly released their experimental dataset, but neither study quantifies the attribution structure underlying predictions.

2.4 Explainable AI and Bankruptcy Prediction

Lundberg and Lee [16] introduced SHAP (SHapley Additive exPlanations), which uniquely satisfies the axioms of local accuracy, missingness, and consistency by mapping model predictions to Shapley values from cooperative game theory. Cho and Shin [17] applied SHAP-weighted counterfactual explanations to bankruptcy prediction, providing actionable remediation paths. Weber et al. [18] confirmed that regulators in highly regulated financial domains now demand traceable transparency in AI-based decisions.

For sequential models, Bento et al. [19] extended KernelSHAP to the sequential domain with TimeSHAP, computing feature-, timestep-, and cell-level attributions for recurrent models. However, TimeSHAP is designed for single-head recurrent architectures and cannot directly decompose head-level contributions in multi-head structures. Quantifying attribution at the head level in multi-head LSTM architectures thus remains an open methodological gap in the literature.

3 Data and Research Model

3.1 Dataset

The dataset comprises annual financial data of U.S. publicly listed companies collected from the Wharton Research Data Services (WRDS) and the Center for Research in Security Prices (CRSP), covering 1999 to 2018. The dataset consists of 8,262 companies with 78,682 observations, publicly

released on GitHub as a benchmark by Pellegrino et al. [9]. Each observation is labeled 1 (bankruptcy) or 0 (non-bankruptcy).

The 18 independent variables span liquidity (current assets, short-term debt), profitability (net income, EBIT, EBITDA, total revenue, net sales, gross profit), leverage (total liabilities, current liabilities, total long-term debt, accounts payable), efficiency (inventory, total receivables, D&A), and scale (total assets, market value, total operating expenses). Data are split chronologically—training (1999–2011), validation (2012–2014), test (2015–2018)—preventing data leakage. Class imbalance (approximately 1:26–1:30) is addressed by undersampling only the training set, preserving realistic imbalance in validation and test sets. Preprocessing uses RobustScaler normalization. Three window sizes (W = 3, 4, 5 years) are tested independently.

Note that the 18 variables contain accounting identity relationships (net_sales ≈ total_revenue; EBITDA = EBIT + D&A), which introduce structural information redundancy. This study follows Pellegrino et al. [9] in retaining the original variable set for benchmark comparability, while acknowledging the potential effect on SHAP attribution distributions.

It should be noted that this study does not directly model transaction-level trade credit behavior, such as payment delays, invoice histories, or customer-supplier network positions. Rather, the proposed framework is intended to support preliminary B2B credit screening when only publicly available financial statements are accessible, especially for evaluating new or insufficiently observed counterparties.

3.2 Multi-Head LSTM Architecture

The adopted MH-LSTM model consists of three sequential components. (1) Multi-Head Layer: Given input tensor (B, W, V)—batch, window, variables—each variable v is assigned a dedicated LSTM head that processes only the univariate series {x_{t,v}} for t = 1,...,W, outputting its final hidden state h_v. This design is intended to reduce the information dilution that can arise when heterogeneous financial variable dynamics are conflated in a single recurrent path. (2) Projection Layer: Each head's hidden state h_v is linearly projected to a lower-dimensional vector, then all V projection vectors are concatenated and normalized via Layer Normalization to unify scale across heads and ensure training stability. (3) Fully Connected Classifier: A Linear–Sigmoid–Linear–Softmax structure converts the normalized representation to binary bankruptcy probability.

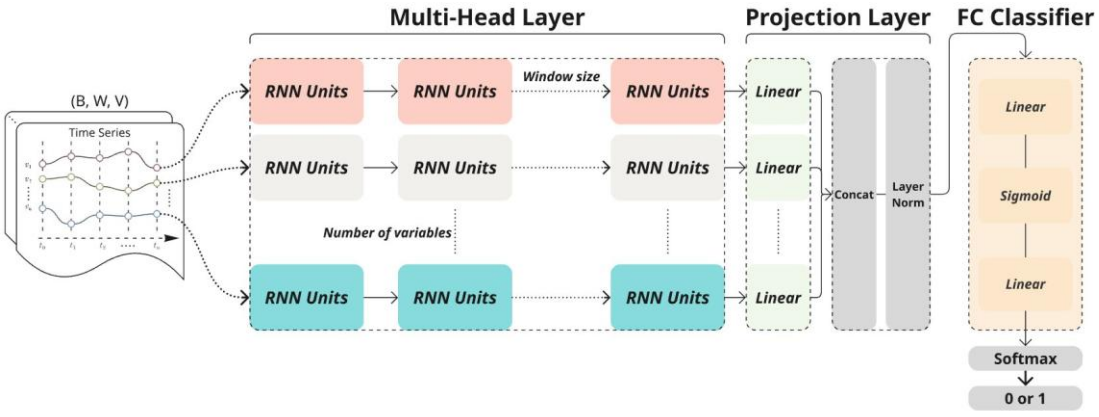


Figure 1. Architecture of the Multi-Head LSTM (MH-LSTM) Model. Each financial variable is assigned an independent LSTM head in the Multi-Head Layer. Projection Layer outputs are concatenated and normalized before being passed to the Fully Connected Classifier.

The multi-head design improves variable-specific temporal representation, but it may also delay or restrict the learning of complex cross-variable interactions during the early temporal feature-extraction stage. In the current architecture, such interactions are mainly captured after the individual head outputs are concatenated and passed to the projection and classification layers.

For fair comparison, total parameters are controlled at approximately 30,000 across all models by adjusting the LSTM hidden size. Training hyperparameters: learning rate 0.0001, batch size 32, maximum 300 epochs with early stopping (patience = 30), CrossEntropyLoss, AdamW optimizer, and ReduceLROnPlateau learning rate scheduler.

To ensure statistical reliability, we adopt a nested evaluation protocol. In each trial, 100 models are trained with different random seeds and their test-set metrics are averaged to yield a single aggregated performance estimate. This procedure is independently repeated for 10 trials, and we report the mean and standard deviation across the 10 trial-level estimates. The standard deviations thus reflect the stability of the aggregated estimator rather than the variability of individual model runs.

3.3 Explainability Methodology

Two complementary XAI methods are applied. Expected Gradients SHAP (GradientExplainer) is selected over TimeSHAP for two reasons. First, TimeSHAP is perturbation-based, incurring prohibitive computational cost for 18-variable multi-period inputs. Second, TimeSHAP is designed for single-head recurrent architectures and cannot naturally decompose head-level contributions in MH-LSTM. The gradient-based Expected Gradients method of Erion et al. [20] estimates SHAP values as:

$$\varphi_i(x) = E_{x' \sim D,\ \alpha \sim U[0,1]}[(x_i - x'_i) \cdot \partial F(x' + \alpha(x - x')) / \partial x_i]$$

This satisfies the local accuracy, missingness, and consistency axioms [16] while efficiently computing per-timestep, per-variable attributions on the 3D input tensor (B × W × V) via backpropagation.

Layer Gradient×Activation (Layer Grad×Act) probes the Projection Layer. The head-level attribution difference is defined as:

$$\Delta_{v,j} = E[Grad \times Act_{v,j} \mid failed] - E[Grad \times Act_{v,j} \mid alive]$$

where $Grad \times Act_{v,j} = (\partial \hat{y} / \partial a_{v,j}) \cdot a_{v,j}$, with $a_{v,j}$ denoting the j-th activation of head v's projection output. Positive $\Delta_{v,j}$ indicates that the corresponding variable head activates more strongly for bankrupt firms in that projection dimension. Both methods share the same backpropagation graph; their convergence is therefore interpreted as internal consistency rather than independent cross-validation. Spearman rank correlation between the two sets of variable importance scores is used to quantify this convergence.

To make the repetition scheme explicit and consistent across the paper, three levels of aggregation are used. The predictive performance metrics in Section 4 follow the nested protocol described above: 100 seeded runs are averaged within each of 10 independent trials, yielding 1,000 trained models in total. The SHAP and Grad×Act importance values in Tables 3–5 are instead averaged over 10 independent runs, while the activation analysis in Figure 4 is drawn from a single representative run. This asymmetry is deliberate: gradient-based attribution over 18-variable multi-period inputs is substantially more expensive than a single performance evaluation, and the figures are intended as qualitative illustrations rather than precise estimates. Accordingly, single-run figure values (e.g., $\rho = 0.936$ in Figure 4) may differ slightly from the corresponding 10-run means reported in the tables (e.g., $\rho = 0.895$ in Table 5). Future work will adopt a single, uniformly repeated protocol across all predictive and attribution analyses, including the figure-level visualizations, so that performance estimates, importance rankings, and their visual summaries are reported on a common repeated basis.

4 Experiments and Results

4.1 Evaluation Metrics

Given severe class imbalance, PR-AUC is adopted as the primary evaluation metric, as it most stringently reflects positive-class (bankrupt firm) detection capability without inflating performance with true negatives. Supporting metrics include ROC-AUC, Balanced Accuracy (BAC), Matthews Correlation Coefficient (MCC), and Macro-F1. For practical B2B credit risk interpretation, bankrupt firm Recall (= 1 − Type 2 Error) and Type 1 Error (healthy firm misclassification rate) are additionally reported. All metrics are reported as mean ± standard deviation across 10 independent trials, where each trial is itself the average over 100 seeded model runs. This two-level design separates estimator variance (across trials) from seed-level stochasticity (within a trial).

4.2 Predictive Performance

ROC-AUC increases monotonically from 0.886 (Window 3) to 0.895 (Window 5), and PR-AUC rises from 0.345 to 0.407, demonstrating that longer financial histories substantively improve bankruptcy detection. Bankrupt firm Recall improves from 0.815 to 0.846, and Type 2 Error (missed bankruptcies) decreases from 0.185 to 0.154. In B2B trade credit risk management, Type 2 Errors translate directly into credit losses; the reduction in missed bankruptcies is therefore practically significant. Precision remains at approximately 0.107 regardless of window size, reflecting the structural constraint of the 1:26 class imbalance combined with undersampling-based imbalance correction. The low precision indicates that the model should not be used as an automatic rejection or approval tool in its current form. Instead, it is more appropriate as a high-recall early-warning filter that identifies firms requiring further human review. In practical B2B credit screening, the decision threshold should be calibrated according to the relative costs of false positives and false negatives, and threshold optimization or cost-sensitive learning may improve the operational trade-off between missed bankruptcies and unnecessary credit restrictions.

Table 1. Test Set Performance Results (mean ± std, across 10 trials; each trial aggregates 100 seeded runs)

Metric	Window 3	Window 4	Window 5
ROC-AUC	0.8861 ± 0.0158	0.8911 ± 0.0094	0.8951 ± 0.0135
PR-AUC	0.3446 ± 0.0357	0.3786 ± 0.0720	0.4073 ± 0.0665
BAC	0.7884 ± 0.0208	0.7978 ± 0.0098	0.7963 ± 0.0137
MCC	0.2383 ± 0.0321	0.2431 ± 0.0256	0.2429 ± 0.0248
Macro-F1	0.5245 ± 0.0337	0.5214 ± 0.0300	0.5201 ± 0.0338
Recall (Bankrupt)	0.8149 ± 0.0260	0.8448 ± 0.0394	0.8461 ± 0.0643
Precision (Bankrupt)	0.1070 ± 0.0203	0.1067 ± 0.0209	0.1074 ± 0.0235
Type 1 Error	0.2381 ± 0.0568	0.2491 ± 0.0493	0.2535 ± 0.0582
Type 2 Error	0.1851 ± 0.0260	0.1552 ± 0.0394	0.1538 ± 0.0643

PR-AUC is consistently higher in the validation set than in the test set, while ROC-AUC shows the opposite pattern. This divergence reflects differences in the bankruptcy composition and

macroeconomic conditions between the validation period (2012–2014) and the test period (2015–2018), reinforcing the importance of reporting multiple complementary metrics rather than relying on a single measure.

Table 2. Validation Set Performance Results (mean ± std, across 10 trials; each trial aggregates 100 seeded runs)

Metric	Window 3	Window 4	Window 5
ROC-AUC	0.8478 ± 0.0103	0.8564 ± 0.0118	0.8492 ± 0.0113
PR-AUC	0.4759 ± 0.0424	0.5064 ± 0.0171	0.4960 ± 0.0183
BAC	0.7690 ± 0.0101	0.7800 ± 0.0150	0.7679 ± 0.0142
MCC	0.3936 ± 0.0251	0.4113 ± 0.0223	0.3895 ± 0.0228
Recall (Bankrupt)	0.7527 ± 0.0506	0.8042 ± 0.0756	0.8117 ± 0.0811
Type 2 Error	0.2473 ± 0.0506	0.1958 ± 0.0756	0.1882 ± 0.0811

4.3 SHAP Feature Importance Analysis

Table 3. SHAP Feature Importance by Window (10-run mean, Top 5 ranked by SHAP)

Rank	W3 Feature	SHAP	W4 Feature	SHAP	W5 Feature	SHAP
1	market_value	0.163	market_value	0.096	market_value	0.092
2	net_income	0.070	net_income	0.093	net_income	0.069
3	total_receivables	0.054	inventory	0.059	current_assets	0.039
4	total_long_term_debt	0.053	total_receivables	0.054	inventory	0.034
5	total_current_liabilities	0.051	total_long_term_debt	0.052	total_receivables	0.029

market_value (market capitalization) consistently ranks first across all three window settings. Its mean SHAP value in Window 3 (0.163) is approximately 2.3 times that of the second-ranked net_income (0.070), and it appears in the top-5 in 8 of 10 experimental runs. As a forward-looking indicator reflecting the collective expectations of market participants, market value incorporates distress signals before they appear in accounting statements, supporting the literature on the predictive superiority of market-based over accounting-based indicators.

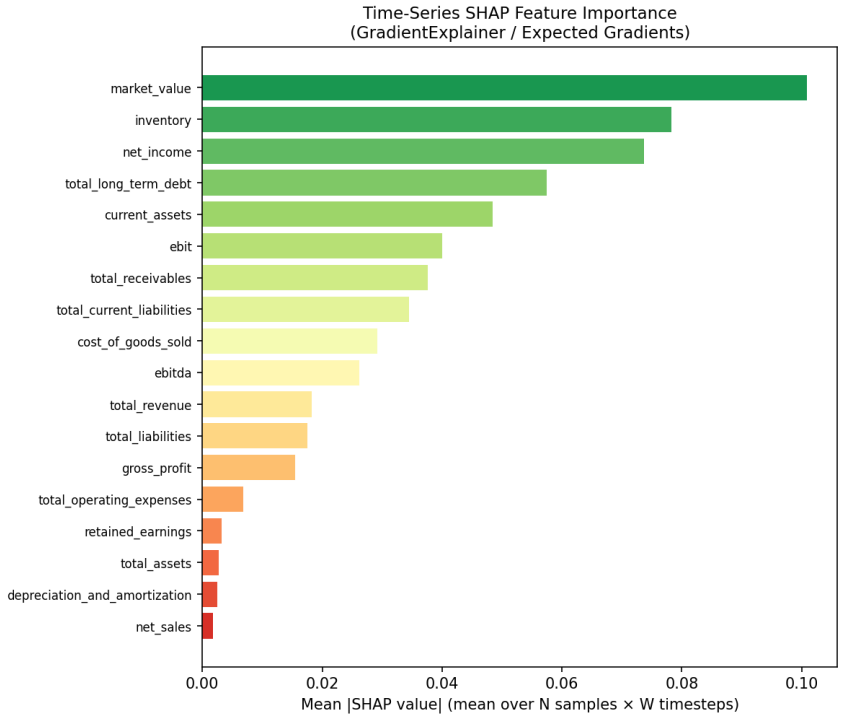


Figure 2. Time-Series SHAP Feature Importance (Window 5, GradientExplainer / Expected Gradients). Mean |SHAP value| averaged over N samples × W timesteps. market_value and net_income consistently dominate, while D&A ranks last across all window settings.

net_income consistently ranks second, confirming that sustained profitability deterioration is a core predictor of bankruptcy. depreciation_and_amortization (D&A) ranks last (18th) across all three windows. As a non-cash expense item, D&A has limited direct linkage to short-term liquidity or credit risk, making this result economically interpretable. net_sales and total_assets also rank near the bottom, partly reflecting information overlap from accounting identity relationships embedded in the dataset.

4.4 Temporal Attribution Analysis

Table 4. Timestep-level SHAP Values for Key Variables, Window 3 (10-run mean)

Variable	t=0 (oldest)	t=1	t=2 (latest)	Ratio (t2/t0)
market_value	0.1124	0.1650	0.2117	1.88×
net_income	0.0311	0.0598	0.1197	3.85×
EBIT	0.0198	0.0362	0.0915	4.62×
total_current_liabilities	0.0241	0.0448	0.0846	3.51×
total_receivables	0.0387	0.0536	0.0697	1.80×

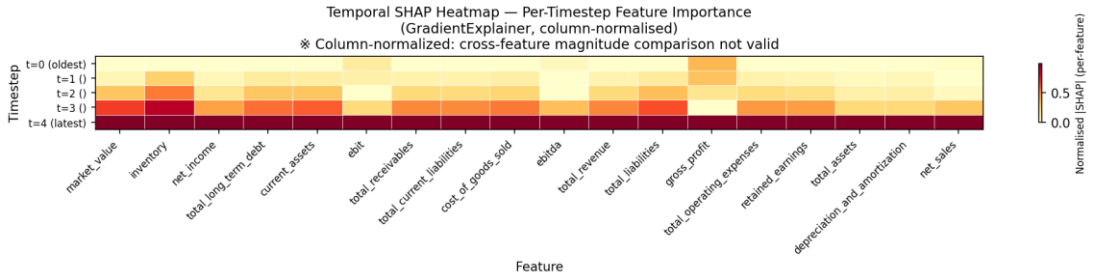


Figure 3. Temporal SHAP Heatmap — Per-Timestep Feature Importance (Window 5, GradientExplainer, column-normalized). Rows represent timesteps t=0 (oldest) through t=4 (latest). Darker color indicates higher normalized SHAP contribution. Major features show a clear recency-weighted attribution pattern.

A consistent monotonic increase—t=0 (oldest) < t=1 < t=2 (latest)—is observed across all variables. The t2/t0 ratios for net_income (3.85×) and EBIT (4.62×) are particularly striking, indicating that recent-year profitability deterioration functions as a far stronger bankruptcy warning than historical records. The same latest-timestep dominance is confirmed in Windows 4 and 5, becoming more pronounced as the window lengthens: as more history becomes available, the model increasingly differentiates between distant baseline information and recent change signals.

This finding has a direct practical implication for B2B trade credit risk management: early warning systems should prioritize timely acquisition of the most recent financial statements. Even with a 5-year window, the most recent year's contribution far exceeds that of earlier periods, making data currency a critical design consideration.

However, this recency-dominant attribution pattern should not be interpreted solely as evidence of financial distress dynamics. It may also partly reflect the inductive bias of LSTM architectures, which tend to preserve and emphasize more recent sequential information through their gating mechanisms. Therefore, future work should compare MH-LSTM with alternative temporal architectures, such as temporal convolutional networks, attention-based models, or Transformers, to distinguish economic recency effects from architecture-driven recency effects.

4.5 Layer Gradient×Activation Analysis and Methodological Convergence

In simple terms, internal consistency means that the two attribution methods point to broadly similar important variables, even though SHAP evaluates importance at the input level whereas Grad×Act evaluates importance inside the learned projection representation.

Table 5. SHAP–Grad×Act Rank Convergence (Spearman ρ, 10-run mean)

Window	Spearman ρ (mean)	Std. Dev.	p-value
Window 3	0.851	0.118	< 0.001
Window 4	0.851	0.130	< 0.001
Window 5	0.895	0.053	< 0.001

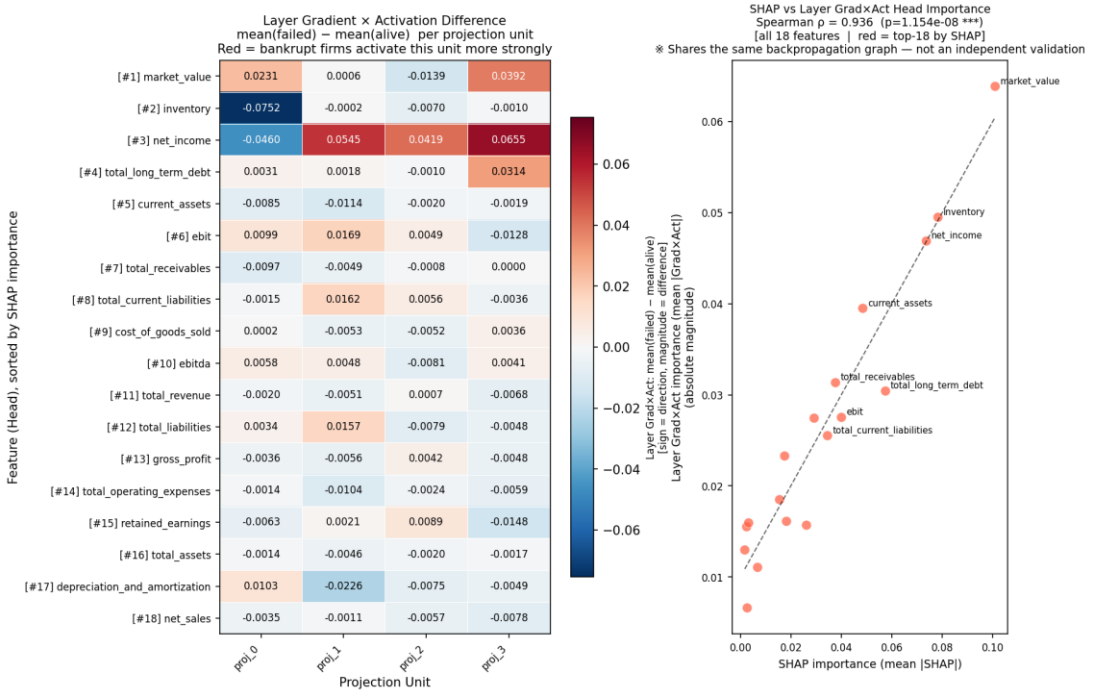


Figure 4. Layer Gradient × Activation Analysis (Window 5). Left: Head-level activation difference ($\Delta_{v,j}$ = mean(failed) − mean(alive)) per projection unit; red = bankrupt firms activate more strongly. Right: Spearman rank correlation between SHAP and Grad×Act importance scores ($\rho = 0.936$, $p = 1.15 \times 10^{-8}$) from a representative single run.

The market_value head shows strong positive activation differences (bankrupt > healthy) in at least one projection unit across all window settings, consistent with its first-place SHAP ranking. The inventory head shows negative activation differences in some projection units, suggesting the model learns that healthy firms maintain more substantial inventory—plausible given that inventory represents working capital and operational continuity. The Spearman rank correlation between SHAP and Grad×Act variable importance rankings is $\rho = 0.851$–0.895 ($p < 0.001$) across all windows, with Window 5 showing the highest convergence ($\rho = 0.895$, std = 0.053).

Since both methods share the same backpropagation graph, their convergence is designated as internal consistency validation rather than independent cross-validation. Nonetheless, the statistical concordance between two analytically distinct perspectives—input-space attribution versus internal representation-space head activation differences—strengthens the reliability of the resulting interpretations.

5 Conclusion and Limitations

5.1 Conclusion

This study adopted an MH-LSTM architecture for corporate bankruptcy early warning using publicly disclosed financial time-series data in B2B trade credit risk contexts, and quantitatively interpreted the prediction basis through Expected Gradients SHAP and Layer Gradient×Activation

analyses.

Four key empirical findings emerge. First, MH-LSTM achieves stable ROC-AUC 0.886–0.895 and PR-AUC 0.345–0.407, with monotonic improvement as the window size increases, demonstrating that longer financial histories substantively enhance distress detection. Second, market_value and net_income consistently rank first and second across all window settings, establishing the predictive superiority of market-based signals and profitability deterioration as primary early-warning indicators. Third, recent fiscal-year financial changes dominate the model's prediction basis, with t2/t0 ratios of 3.85× (net_income) and 4.62× (EBIT) in Window 3, implying that B2B credit risk systems should prioritize real-time financial data acquisition. Fourth, the Spearman ρ = 0.851–0.895 convergence between SHAP and Grad×Act provides internal consistency evidence supporting the interpretability of the proposed framework.

5.2 Limitations and Future Research

Several limitations merit acknowledgment. First, accounting identity relationships among the 18 variables introduce structural information redundancy that may affect SHAP attribution distributions; future work should systematically examine the impact of variable selection on XAI stability. Second, the shared backpropagation graph of SHAP and Grad×Act precludes fully independent cross-validation; perturbation-based or permutation-based methods should be compared for external validation. Third, the dataset is confined to U.S. publicly listed firms over 1999–2018. Although this period includes major macroeconomic events such as the 2008 financial crisis, the model does not explicitly incorporate macroeconomic indicators, industry-level shocks, or credit-cycle variables. This limits the ability to separate firm-specific distress signals from broader economic-cycle effects. Moreover, the findings may not generalize directly to private SMEs, whose B2B trade credit risk is often more volatile and less transparent due to limited disclosure, weaker reporting standards, and sparse market-based information. Fourth, Precision remains at approximately 0.107 regardless of window size, reflecting the intrinsic constraint of undersampling-based imbalance correction; alternative strategies (SMOTE, cost-sensitive learning, threshold optimization) warrant systematic evaluation. Fifth, model capacity effects and architectural alternatives (attention mechanisms, Transformer-based structures) remain unexplored.

Because the primary objective of this conference paper is to interpret the prediction basis of an existing MH-LSTM framework rather than to establish its superiority over all competing models, we do not conduct a full benchmark comparison against single-head LSTM, XGBoost, or LightGBM under identical experimental settings. Future research should directly compare MH-LSTM with strong sequential and non-sequential baselines to isolate the marginal utility of the multi-head design.

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